

1(a) OCR Insurance uses a computer system to calculate the price that customers pay for car insurance.

The computer system contains a CPU, GPU, RAM and ROM.

i. State **two** factors that affect the performance of a CPU.

1 _____

2 _____

[2]

ii. Explain the difference between RAM and ROM, including how these are used by the computer system.

[4]

iii. Describe **one** non-graphical use for a GPU.

[2]

- (b) The CPU uses pipelining to improve efficiency.

Explain what is meant by the term 'pipelining'.

[3]

- (c) * OCR Insurance's computer system uses secondary storage across the company in servers, client machines and for backup purposes.

For each of these, discuss whether magnetic storage or solid state storage would be most suitable, taking into account the advantages and disadvantages of both.

[illegible]

[9]

<u>CustomerID</u>	Surname	Title	Phone	CarReg
JJ178	James	Mr	(0121) 343223	DY51 KKY
HG876	Habbick	Miss	(01782) 659234	PG62 CRG
EV343	Elise	Mrs	(07834) 123998	HN59 GFR
PG127	Pleston	Mr	(07432) 234543	JB67 DSF

<u>CustomerID</u>	Surname	Title	Phone	CarReg
JJ178	James	Mr	(0121) 343223	DY51 KKY
HG876	Habbick	Miss	(01782) 659234	PG62 CRG
EV343	Elise	Mrs	(07834) 123998	HN59 GFR
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[2]

-----[4]

- iii. Describe **one** problem that would arise with the flat file database structure if a customer wanted to insure more than one car at the same time.

-----[2]

- iv. Describe how the flat file database structure could be altered to efficiently allow each customer to insure multiple cars at the same time. (You may assume each car is insured to only one customer.)

-----[5]

2(a)

i. Convert the denary number 231 to an unsigned 8-bit binary number.

[1]

ii. Convert the hexadecimal number 6F to an unsigned 8-bit binary number.

[1]

(b) The floating point number 001101 0100 is stored using 6 bits for the mantissa and 4 bits for the exponent, both in two's complement. This number is not normalised.

i. Give the normalised version of this number, showing your working.

[4]

- ii. Convert your answer to part (i) to denary, showing your working.

[3]

- $$\begin{array}{r} 0110000110 \\ + 0101000100 \\ \hline \end{array}$$

[5]

3(a) A website sells tickets for sporting events. The website uses HTML, CSS and JavaScript.

Describe the purpose of HTML and CSS within the code of the website.

HTML

CSS

- [4]
- (b) One page in the website contains a hyperlink on an image. When the image stored as “ticket.png” is clicked, the user is hyperlinked to the page stored as “booking.htm”.

Write the HTML code to implement this hyperlink.

[3]

- (c) A website charges a booking fee of £2.99 on each ticket sold. In addition, if the tickets are purchased from outside of the UK, £4.99 is added to the booking fee. The booking fee is calculated using a JavaScript function named `bookingfee()`.

Complete the definition of the `bookingfee()` function below.

```
function bookingfee(numtickets, country) {  
  
    var nonUKprice = 4.99;  
  
    var perTicketPrice = .....;  
  
    var total = 0;  
  
    if (country!="UK") {  
  
        total = total + .....;  
  
    }  
  
    total = total + (..... * perTicketPrice);  
  
    ..... total;  
  
}
```

[4]

(d) The JavaScript function above is used to show users the booking fee. When users click to buy the tickets, the booking fee is calculated again on the server.

i. Explain why server side processing is used to recalculate the booking fee.

[3]

ii. Explain **one** advantage of client side processing to either the customer buying the tickets, or to company who own the website.

[2]

(e) Users are able to search for and find the ticket website using a search engine. Search engines can use indexing and ranking.

i. Describe how a website is indexed by a search engine.

[4]

ii. A search engine can use the PageRank algorithm to determine a website’s ranking. The PageRank algorithm utilises a damping factor.

State what is meant by the term ‘damping factor’.

[1]

iii. Give **two** other factors that affect the output value given by the PageRank algorithm for a website.

1 -----

2 -----

[2]

4 * *“The Computer Misuse Act means that computer users are criminalised for simply trying to learn how systems work.”*

Discuss whether or not you agree with this statement.

----- [9]

- 5(a) All users of a computer system have a unique username and password. The computer system has implemented two-factor authentication so that users must respond to either an email or text message containing a secret code to be able to access the system.

Let:

A be a Boolean value for if a user enters a valid username

B be a Boolean value for if a user enters a password that matches their username

C be a Boolean value for if a user is able to respond to an email containing a secret code

D be a Boolean value for if a user is able to respond to a text message containing a secret code

Q be a Boolean value for if entry to the computer system is allowed

Complete the Boolean expression below:

$Q \equiv$ -----

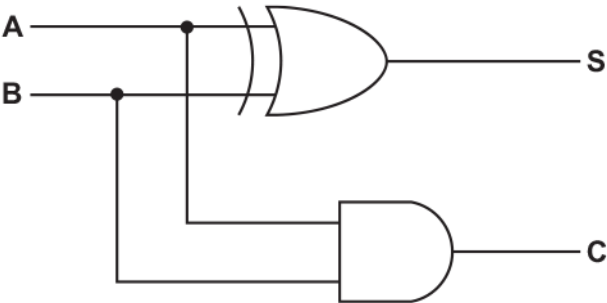
- (b) Another Boolean expression for a logic system is shown below:

$$Q \equiv \neg (\neg A \wedge \neg B)$$

- i. Simplify this Boolean expression so that it does not include any negation. You must explain which Boolean algebra rule(s) you are using at each step.

----- [2]

(c)



The logic circuit above has two inputs (A, B) and two outputs (S, C).

i. Give the Boolean expressions for the outputs S and C.

S ≡ _____
C ≡ _____

[2]

ii. Complete the truth table for this logic circuit.

A	B	S	C
0	0		
0	1		
1	0		
1	1		

iii. Describe how this logic circuit can be adapted to add together two 4-bit binary numbers.

[4]

6(a) A program written using the Little Man Computer instruction set is shown in Fig. 1.

```

      INP
      STA numone
      INP
      STA numtwo
main  LDA numone
      SUB numtwo
      BRP pos
notpos LDA count
      OUT
      LDA numone
      OUT
      HLT
pos   STA numone
      LDA count
      ADD one
      STA count
      BRA main
numone DAT
numtwo DAT
one    DAT 1
count  DAT 0
```

Fig. 1

various registers are used when the program above is executed.

i. State what is meant by the term ‘register’.

[2]

ii. Explain how the accumulator is used when the line BRP pos is executed.

[2]

(b) Complete the table below to show the output(s) from this program given the inputs.

Inputs	Output(s)
12, 5	
18, 2	
16, 4	
3, 7	

[4]

(c) Write an algorithm using pseudocode that has the same functionality as the code in Fig. 1.

Blank lined paper for writing.

(d) * In assembly language, different modes of addressing memory can be used.

Discuss the different modes used. You should include:

- How the operand value is determined
- What an operand of 27 would refer to in that mode
- The reasons for requiring multiple modes of addressing

[12]

7(a)

A business uses an array with the identifier wNames to store workers' names. A variable with the identifier top is used to store the index of the last element to be added to the array, which is also the element which will next be removed.

wNames						
0	1	2	3	4	5	6
Kirstie	Martyn	Louise	Alex	Anna		

top

4

i. State the name of the type of data structure described above.

[1]

ii. Using pseudocode, write an algorithm that allows the user to enter a name which is then pushed onto the data structure above, checking first that the data structure is not full.

[4]

(b) The same workers' names are stored in a binary search tree which is ordered alphabetically.

Kirstie is set as the root node, with Martyn, Louise, Alex and Anna added one by one.



i. Complete the tree diagram above to show where Martyn, Louise, Alex and Anna would be added to this binary search tree.

[4]

ii. Describe the process of using the binary search tree above to search for the name "Zoe".

[3]

iii. Compare the efficiency of a binary search tree to a linked list when searching for data.

[2]

iv. Compare the efficiency of a binary search tree to a hash table when searching for data.

[2]

(c) An object oriented system is implemented to organise further information about each worker's attendance. Classes, objects, methods and attributes are used in this system.

i. State the meaning of each of the following terms:

Object -----

Method -----

Attribute -----

[3]

ii. Each worker has a name and an attendance figure which can be between 0 and 100.

Write a definition for a fully encapsulated worker class, providing both get and set methods for all attributes. You do **not** have to write code for the constructor method.

[5]

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1	a	i	• Clock speed • Number of cores • Cache	2 AO1.1	Accept Use of pipelining/size of pipeline Use of out of order execution Use of SIMD instructions Integrated graphics processing on CPU
		ii	• RAM is <u>volatile</u> • Used for storing programs/data/parts of OS <u>currently in use</u> • ROM is <u>non volatile</u> • Used for storing (e.g.) BIOS / bootstrap	4 AO1.2	
		iii	• Performing complex numerical calculations • Calculations on matrices / vectors / multiple data at the same time • ...e.g. insurance pricing, modelling risk, calculating bills	2 AO2.2	Example has to relate to insurance company
	b		• Concurrent processing of multiple instructions • One instruction can be fetched while previous is being decoded... • And the one before is being executed. • In case of a branch pipeline is flushed. • Increases speed of execution	3 AO1.1	
	c		Mark Band 3–High Level (7-9 marks) The candidate demonstrates a thorough knowledge and understanding of magnetic and solid state storage. The material is generally accurate and detailed. The candidate is able to apply their knowledge and understanding directly and consistently to the context provided. Evidence/examples will be explicitly relevant to the explanation. The candidate provides a thorough discussion which is well balanced. Evaluative comments are consistently relevant and well-considered. There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated. Mark Band 2–Mid Level (4-6 marks) The candidate demonstrates reasonable knowledge and understanding of magnetic and solid state storage; the material is generally accurate but at times underdeveloped.	AO1.1 (2), AO1.2 (2), AO2.1 (2), AO3.3. (3)	Data stored by altering the magnetic field of a small section of the tape or disk; one orientation represents a binary value of 1, the opposite orientation represents a binary value of 0. Written to using a read head. Read head can then read the magnetic field and retrieve the data. These read and write heads move across the disk or the tape moves under the heads. Solid state devices store data using semiconductors; drives can be based on NAND Flash or DRAM. No moving parts are required. Backup involves copying potentially large amounts of data to and from storage devices at regular intervals. Servers distribute data to client devices. AO2 Magnetic hard drives entirely suitable for storing operating system, software and data. Large capacity, cheap (by comparison) purchasing price and reliability make them ideal for heavy use in an insurance office. Solid state drives have no moving parts

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
			The candidate is able to apply their knowledge and understanding directly to the context provided although one or two opportunities are missed. Evidence/examples are for the most part implicitly relevant to the explanation. The candidate provides a sound discussion, the majority of which is focused. Evaluative comments are for the most part appropriate, although one or two opportunities for development are missed. There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence. Mark Band 1-Low Level (1-3 marks) The candidate demonstrates a basic knowledge of magnetic and solid state storage; the material is basic and contains some inaccuracies. The candidate makes a limited attempt to apply acquired knowledge and understanding to the context provided. The candidate provides a limited discussion which is narrow in focus. Judgments if made are weak and unsubstantiated. The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear. 0 marks No attempt to answer the question or response is not worthy of credit.		and offer an increase in speed and quicker access to customer records. However, smaller capacity (in comparison) may be an issue for larger data sets. Backup systems may be required to maximise storage whereas access speed is not so much of an issue. Limited lifespan of SSDs due to limited read/writes may restrict use in data heavy applications. Servers will require quick access to data to serve multiple client machines but will also be extremely read/write heavy. AO3 Both suitable for the insurance office but for different purposes. SSDs to store operating system and program files will ensure quicker boot up and loading of programs. May be useful for both client and servers to store operating systems. Magnetic hard drives to store large quantities of data provides reliability and large capacity at small (by comparison) cost. May be most suitable for storing large amounts of data on client devices and for backup systems. Magnetic tape device may be suitable for backing up customer data as too slow for use on day-to-day basis. Use of solid state devices for storing customer data that is frequently accessed means needing to plan for replacement of drives when read/write limit reached. Solid state devices better if used in mobile devices (e.g. for mobile insurance workers) as relatively small chance of damage if dropped as no moving parts.
	d	i	- Field with a unique value - CustomerID	2 AO1.2 (1), AO2.2 (1),	
		ii	- SELECT CustomerID, Surname - FROM Customer - WHERE Title="Miss" - OR Title = "Mrs"	4 AO3.2	1 mark per bullet point. Data in fields must be in speech marks/apostrophes. Allow speech mark/apostrophe FT for BP 2 and 4. Ignore colons/semicolons.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		iii	• Only one customer entry allowed (because of key field) • ...so would not be able to add second entry • Customer data already present/would be repeated •resulting in redundant data/wasted space • ...resulting in inconsistencies should changes be made	2 AO2.1	Mark in pairs.
		iv	• Add in second table • ...for the cars // splitting up cars/customers • The primary key of customer is used as a field in the car/vehicle table • ... as a foreign key of cars • Create one to many relationship. <i>Accept many to one.</i>	5 AO1.2	
			Total	33	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
2	a	i	11100111	1 AO1.2	
		ii	0110 1111	1 AO1.2	
	b	i	- Move decimal place 1 place right = 0.1101 to normalise - Pad with 0s, <u>mantissa is 011010</u> - Exponent is 4 / Subtract 1 from exponent... - New exponent is 3 = <u>0011</u>	4 AO1.2	
		ii	- move dp 3 places right 0110.1 6.5	3 AO1.2	Accept conversion of un-normalised number (will result in same output) Accept method of using 0.1101×2^3 FT required from part (i) if this is done incorrectly.
	c		- Calculating exponent of 6 for first number/Calculating exponent of 4 for second number - Moving binary point correct number of places in both mantissa to give 0110000.0 and 01010.0 - Addition of 0110000.0 and 01010.0 to give 0111010.0 - Movement of binary point six places to left to match original mantissa format and give value of 011101 - Calculate exponent in original format to give value of 0110	5 AO1.2	Alternative: - Calculation of exponents (6 and 4) // difference between exponents of 2 - Move <u>first mantissa 2 places left</u> // <u>second mantissa 2 places right</u> to match exponents ...giving 011.000 + 0.10100 (exp 4) // 0.11000 + 0.00101 (exp 6) - Correct addition (answer of 011.1010 // 0.111010) - Final answer normalised (011101 exp 6) If exponents initially matched to 4, final bullet point should only be given if answer is normalised. If exponents matched to 6, answer will already be normalised. Ensure that answer is in format described (6 bits for mantissa 4 bits for exponent) to gain final bullet point.
			Total	14	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
3	a		- HTML defines the structure of a web page - HTML defines the content of a web page - Using tags (enclosed in <>) - CSS defines the style / appearance - Using selectors such as classes / IDs / etc - Can be placed within HTML or externally in a file - Multiple pieces of CSS can be combined (the more local instances overriding)	4 AO1.1	Do not accept layout/format for HTML as this is too vague and can be used to describe CSS
	b		- Correct <a> with close - href property to correct page - correct tag with src property to correct file	3 AO3.2	
	c		2.99 - nonUKprice - numtickets - return	4 AO3.2	Correct answer only. Penalise spelling if incorrect. Do not accept £ sign in first bullet point.
	d	i	- processing done away from the user's control/Client side processing could be altered - Browser may not support client-side language/ scripting could be turned off - Booking fee calculation needs to be correct for all locations	3 AO2.1	
		ii	To the customer any 2x1: - No need to submit to server and wait for response - Website will work more quickly for user To the company any 2x1: - Reduces load on the server - Will need to spend less on processing power/bandwidth	2 AO1.2	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	e	i	• Web crawler /spider visits site • Either be selecting it from an existing list or following a link. • Records information • ...such as text / metatags / etc • Records the position of each word within the page • Storing them in an index • Follows links to other sites • Robots.txt file can be used to instruct web crawlers	4 AO1.1	
		ii	• A value between 0 and 1 • Probability that a user will not follow a link	1 AO1.1	
		iii	• Number of links to target site • PageRank score of those sites (linking inwards)	2 AO1.1	
			Total	23	

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
4	Mark Band 3–High Level (7-9 marks) The candidate demonstrates a thorough knowledge and understanding of legislation including the Computer Misuse Act. The material is generally accurate and detailed. The candidate is able to apply their knowledge and understanding directly and consistently to the context provided. Evidence/examples will be explicitly relevant to the explanation. The candidate provides a thorough discussion which is well balanced. Evaluative comments are consistently relevant and well-considered. There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated. Mark Band 2-Mid Level (4-6 marks) The candidate demonstrates reasonable knowledge and understanding legislation including the Computer Misuse Act; the material is generally accurate but at times underdeveloped. The candidate is able to apply their knowledge and understanding directly to the context provided although one or two opportunities are missed. Evidence/examples are for the most part implicitly relevant to the explanation. The candidate provides a sound discussion, the majority of which is focused. Evaluative comments are for the most part appropriate, although one or two opportunities for development are missed. There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence. Mark Band 1-Low Level (1-3 marks) The candidate demonstrates a basic knowledge of legislation including the Computer Misuse Act; the material is basic and contains some inaccuracies. The candidate makes a limited attempt to apply acquired knowledge and understanding to	9 AO1.1 (2), AO1.2 (2), AO2.1 (2), AO3.3. (3)	AO1 Computer Misuse Act is legislation aimed at criminalising unauthorised access to a computer system Three stages: Unauthorised access to a computer system Unauthorised access with intent to commit further offences Unauthorised modification of computer material Punishable by up to twelve months in prison and an unlimited fine. AO2 Computer users who investigate how systems work require authorisation in order to not break the Act. Examples such as changing a social media post on a friend's mobile phone potentially breaks all three sections of the Act. Investigation of systems can break the Act without intent, e.g. by changing server logs because of their actions. Users must be aware of the Act (as with any other law) in order to be responsible. AO3 Material available online (e.g. self study videos) that explain how systems work and teach without the need to investigate using unauthorised access. Investigating systems that you own yourself or have authorisation to access does not break the law. Systems are offered to users with strict conditions attached and investigation is not a legitimate excuse for breaking the law. Ethical / white hat hackers will not break this law because they have authorisation. Grey and black hat hackers will break Computer Misuse Act.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
			the context provided. The candidate provides a limited discussion which is narrow in focus. Judgments if made are weak and unsubstantiated. The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear. 0 marks No attempt to answer the question or response is not worthy of credit.		
			Total	9	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance																				
5	a		$Q \equiv A \wedge B$ $\wedge$ $(C \vee D)$	3 AO1.2	$Q \equiv A \wedge B \wedge (C \vee D)$ Accept alternative symbols for AND / OR e.g. $Q = A \text{ AND } B \text{ AND } (C \text{ OR } D)$ Brackets must be included for 3 rd point Allow XOR for bullet point 3 Any additional symbols max 2 marks																				
	b		• Identification of De Morgan's and/or double negation rule • Correct final answer to give $A \vee B$	2 AO2.2																					
	c	i	$S = A \vee B$ $C = A \wedge B$	2 AO2.1	Accept alternative symbols for AND /XOR																				
		ii	• 1 mark for S column • 1 mark for C column	2 AO2.1	<table><tr><th>A</th><th>B</th><th>S</th><th>C</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr></table>	A	B	S	C	0	0	0	0	0	1	1	0	1	0	1	0	1	1	0	1
A	B	S	C																						
0	0	0	0																						
0	1	1	0																						
1	0	1	0																						
1	1	0	1																						
		iii	• Logic circuit adds together 2 binary digits / half adder • S gives sum, C gives carry • Two half adders can be joined together... • ...with an OR gate • to form full adder • 4 full adders can be used to add two four bit numbers • Carry out on one joined to carry in on next	4 AO1.2																					
			Total	13																					

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
6	a	i	• Temporary storage/memory location... • ...inside the CPU • Used for a single specific purpose • Faster access speed than RAM / secondary storage	2 AO1.1	
		ii	• Accumulator checked to see if value held is positive or <u>zero</u> • If so, BRANCH carried out / jumps to specified location.	2 AO1.2	
	b		• 2, 2 • 9,0 • 4,0 • 0,3	4 AO2.2	Code finds integer division and remainder.
	c		• Input two numbers <u>into two separate variables / other suitable data structure</u> • Correctly calculate integer division • Correctly calculate remainder • Print out both	4 AO3.2	Can be completed either by using MOD / DIV or by using repeated subtraction as in LMC example

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
	d	Mark Band 3–High Level (9-12 marks) The candidate demonstrates a thorough knowledge and understanding of different modes of addressing. The material is generally accurate and detailed. The candidate is able to apply their knowledge and understanding directly and consistently to the context provided. Evidence/examples will be explicitly relevant to the explanation. The candidate provides a thorough discussion which is well balanced. Evaluative comments are consistently relevant and well-considered. There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated. Mark Band 2-Mid Level (5-8 marks) The candidate demonstrates reasonable knowledge and understanding of different modes of addressing; the material is generally accurate but at times underdeveloped. The candidate is able to apply their knowledge and understanding directly to the context provided although one or two opportunities are missed. Evidence/examples are for the most part implicitly relevant to the explanation. The candidate provides a sound discussion, the majority of which is focused. Evaluative comments are for the most part appropriate, although one or two opportunities for development are missed. There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence. Mark Band 1-Low Level (1-4 marks) The candidate demonstrates a basic knowledge of different modes of addressing; the material is basic and contains some inaccuracies. The candidate makes a limited attempt to apply acquired knowledge and understanding to the context provided.	12 AO1.1 (2), AO1.2 (2), AO2.1 (3), AO3.3. (5)	AO1 Immediate addressing is where the operand holds the actual data to be used; no address referenced Direct addressing is where the operand holds the address that holds the data to be used. Indirect addressing is where the operand holds an address which is where the data to be used is stored Indexed addressing is where the operand holds an address which is offset using the Index Register to find the true address of the data to be used AO2 Immediate addressing; operand of 27 would refer to the value 27 Direct addressing; operand of 27 would tell the processor to fetch the data held at address 27. Indirect addressing; operand of 27 would tell the processor to fetch the data held at address 27, which itself would then be used an address to fetch data from. Indexed addressing; operand of 27 would be added to the value of the Index register and this would then be used as the address to fetch data from. AO3 Immediate addressing allows simple access to data with no fetch required, but limited by the data size of the operand. Direct addressing allows data to be fetched from memory. Data can potentially be larger in size that with immediate addressing but address range limited by size of operand. Indirect addressing allows a larger range of addresses to be accessed as address fetched. However, multiple fetches required to access data. Indexed addressing allows the Index register to be manipulated to access data stored sequentially e.g. in an array.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
			The candidate provides a limited discussion which is narrow in focus. Judgments if made are weak and unsubstantiated. The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear. 0 marks No attempt to answer the question or response is not worthy of credit.		
			Total	24	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
7	a	i	Stack	1 AO1.1	Correct answer only
		ii	- Input name from user - Check if stack is full e.g.(top <=5) - If not, update top pointer - Correctly push name to index referenced by top	4 AO3.2	<u>Example answer</u> new = input("enter a name : ") if top <=5 then top = top + 1 wNames[top] = new end if
	b	i	1 mark per name inserted in correct place in diagram	4 AO2.1	<pre> graph TD Kirstie --> Alex Kirstie --> Martyn Alex --> Empty1[] Alex --> Anna Martyn --> Louise Martyn --> Empty2[] </pre>
		ii	- Compare with Kirstie – Zoe is larger so go right - Compare with Martyn - Zoe is larger so go right - No right element so stop/not found	3 AO2.2	Allow FT from b(i)
		iii	- Binary tree more efficient than linked list - Do not need to check every value / tree removes half values each time	2 AO1.2	Allow reference to big O for second mark. Linked List O(n), Binary Tree O(log n)
		iv	- Binary tree less efficient than hash table - Hash table can find data immediately / without checking other values.	2 AO1.2	Allow reference to big O for second mark. Binary tree O(log n) Hash table O(1)
	c	i	- Object – instantiated from class - Method – action object performs / link to procedure/functions - Attribute – value held by object / link to variable	3 AO1.1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		ii	• Class definition statement • Defining name and attendance attributes • Appropriate get methods for name <u>and</u> attendance that return a value and have no parameter • Appropriate set methods for name <u>and</u> attendance that take a parameter • ...that restricts attendance to be 0 to 100.	5 AO3.2	<u>Example answer</u> <pre> class Worker private name private attendance public function getName() return name end function public function getAttendance() return attendance end function public procedure setName(newName) name = newName end procedure public procedure setAttendance(newAttend) if newAttend >=0 and newAttend <=100 then attendance = newAttend end if end procedure end class </pre>
			Total	24	